SYNOPSIS

on

PAINT PLANT INTELLIGENCE SYSTEM

(IoT Based Smart Industrial Environment Monitoring System for Paint Plants)

Bachelor of Technology - Electronics & Computer Science Department
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1. Introduction

The Internet of Things (IoT) allows sensors and controllers to collect data and take decisions automatically. Paint and coating plants handle flammable and harmful solvents, and their working conditions (temperature, humidity and air quality) affect both worker safety and the quality of the paint finish. This project proposes the Paint Plant Intelligence System, a low-cost IoT system built around an ESP32 microcontroller that continuously monitors these conditions, warns workers using LEDs and a buzzer, and shows live status on a local Wi-Fi dashboard.

2. Problem Statement

In many paint manufacturing and coating units, the environment is checked manually and only at intervals. Solvent vapours can build up in poorly ventilated areas, temperature can rise unnoticed, and high humidity can spoil the drying of paint. Commercial monitoring systems are expensive, and many low-cost IoT systems depend on internet and cloud services. There is a need for an affordable system that monitors temperature, humidity and solvent vapour / air quality, gives immediate warnings and works locally without internet.

3. Objectives

1. To design a low-cost IoT-based environment monitoring system for paint plants.
2. To monitor temperature, humidity and solvent vapour / air quality continuously.
3. To compare readings with predefined safety limits and decide automatically whether conditions are safe.
4. To give visual (green / red LED) and audible (buzzer) alerts.
5. To treat a sensor failure as unsafe so that the system fails safe.
6. To provide a local Wi-Fi web dashboard that works without internet and can be viewed on a phone or laptop.

4. Scope of the Project

The project is a working prototype for areas such as paint mixing rooms, spray booths, solvent storage and drying zones. It covers sensing, decision making, local alerts and a local dashboard. Features such as calibrated gas measurement, cloud storage and mobile notifications are kept as future enhancements.

5. Literature Survey

Existing work on ESP32-based monitoring systems uses sensors such as DHT11/DHT22 and MQ-series gas sensors and usually sends the data to cloud platforms such as ThingSpeak or Blynk [1]-[3]. Some recent work has even used embedded neural networks on the ESP32 for gas monitoring in industrial safety [4]. The limitations of these systems are that many depend on internet and cloud services, are meant for general air-quality or weather monitoring rather than a specific industry, or are more complex to build. The proposed system focuses on the paint plant and works locally, with its own alerts and dashboard.

6. Proposed System and Methodology

The system has four stages: (1) **Data collection** - the DHT11 measures temperature and humidity and the MQ-135 module gives a digital gas-detected signal; (2) **Data processing** - the ESP32 reads the sensors every 2 seconds and compares them with the safety limits; (3) **Automatic action** - the green LED is ON in safe condition, and the red LED and buzzer are activated in a hazard condition; (4) **Monitoring** - the ESP32 creates its own Wi-Fi network and hosts a web dashboard that refreshes every 3 seconds.

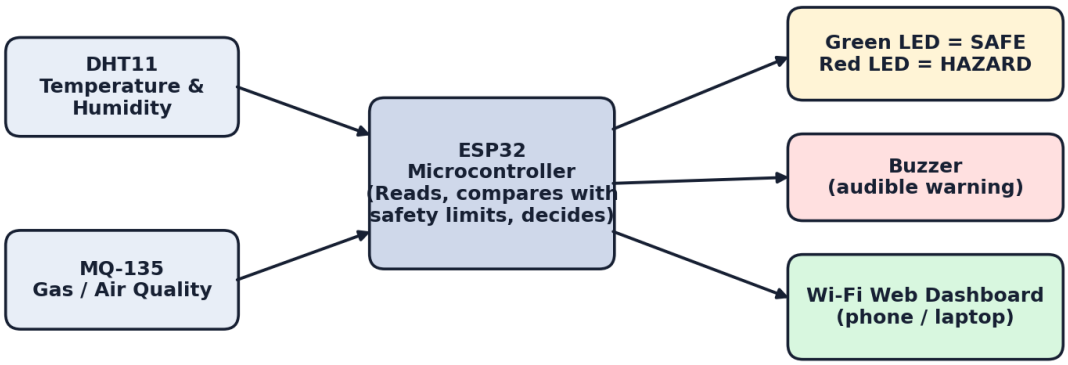


Fig 1. Block diagram of Paint Plant Intelligence System

Safety limits used: temperature above 40 °C, relative humidity above 85 %, and gas detected by the MQ-135 digital output. An invalid DHT11 reading is also treated as a hazard.

7. Hardware and Software Requirements

Component	Quantity	Purpose
ESP32 DevKit V1 (30 pin)	1	Main controller and web dashboard host
DHT11 sensor	1	Temperature and humidity measurement
MQ-135 gas sensor module	1	Solvent vapour / air-quality indication
Green and red LEDs	1 each	Safe / hazard indication
Active buzzer	1	Audible warning
220 Ω resistors, breadboard, jumper wires	As required	Circuit assembly

Software: Arduino IDE, C++ (Arduino framework), DHT sensor library by Adafruit, ESP32 WiFi and WebServer libraries, HTML and CSS for the dashboard.

8. Expected Outcomes and Applications

- Continuous real-time monitoring of temperature, humidity and gas status.
- Immediate LED and buzzer warning when any safety limit is crossed.
- Remote viewing of the plant status from a phone or laptop without internet.
- Applications: paint mixing rooms, spray booths, solvent storage, drying and curing areas, small paint shops, and other chemical or coating industries.

9. Limitations

This is a prototype and not a certified industrial safety device. The MQ-135 digital output depends on its adjustable threshold and calibration and does not give gas concentration in PPM. The DHT11 has limited accuracy.

10. Future Scope

Calibrated gas sensors, automatic exhaust-fan control, data logging, SMS or mobile-app notifications, dew-point-based painting guidance, trend-based early warning, and multi-zone monitoring.

11. Conclusion

The Paint Plant Intelligence System offers a simple, affordable and expandable solution for monitoring the working environment of a paint plant. By combining sensing, automatic decisions, local alerts and a local dashboard, it can help improve worker safety and maintain suitable conditions for paint application.

12. References

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[4] T. P. T. Tran and T. T. Vu, "Development of an IoT-Based Gas Monitoring System Using Artificial Neural Networks on ESP32 for Industrial Safety Applications," JMST, doi: 10.65154/jmst.1035.
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